



Synthesis and Characterisation of Eco-Friendly Chitin-Based Superabsorbent Polymers for Fertiliser Slow Release, Water Absorption Kinetics, and Plant Growth Enhancement

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Abstract

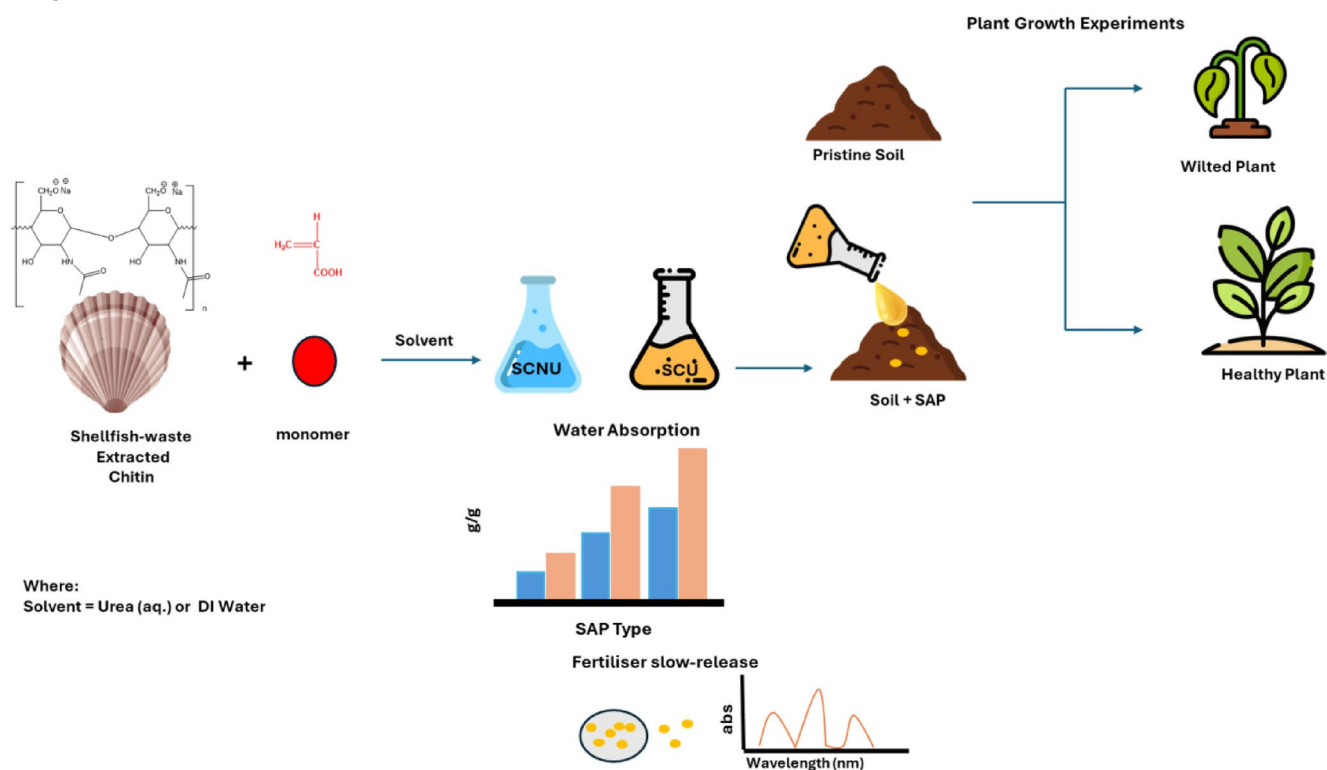
With the global agricultural sector facing major challenges including; drought, water runoff, and fertiliser loss, the growing demand for sustainable practices has intensified. As a result, superabsorbent polymers (SAPs) have garnered attention due to their ability to improve soil water retention and the controlled, slow-release of nutrient molecules. This study explores the synthesis and characterisation of semi-synthetic chitin-based SAPs derived from shellfish waste, offering an eco-friendly alternative to conventional synthetic SAPs. Two SAP variants were synthesised: one without urea (SCNU) and the other incorporating urea (SCU) to enhance solubility and performance. The chemical structure, absorption kinetics, and urea slow-release behaviour were analysed using Fourier-transform infrared (FTIR) spectroscopy, zeta potential measurements, and UV/Vis spectrophotometry. Results revealed that the urea-based SAP (SCU) exhibited superior water absorption and swelling capacity, reaching a maximum absorption capacity of 355 g/g compared to 155 g/g for SCNU SAP. Both SAPs exhibited good biodegradability within just 7 days, highlighting their environmental benefit. We also investigated the SAPs' surface charge behaviour across a range of pH values by measuring zeta potential, which provided insight into their water absorption capacity in varying environmental conditions. Absorption experiments supported the trends observed in the zeta potential analysis. As pH increases, the surface charge becomes more negative, which enhances electrostatic repulsion between polymer chains and results in greater water absorption. Desorption studies showed sustained water release over 14 days, indicating potential to reduce irrigation frequency. Post-synthetic urea loading confirmed the SAPs' slow-release fertiliser capability. Incorporation of indole butyric acid (IBA) further enhanced plant growth, demonstrating the SAPs' dual function in moisture retention and nutrient delivery.

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Graphical Abstract



Keywords Sustainable agriculture · Chitin · Superabsorbent polymers · Water retention · Slow-Release fertiliser

Introduction

The agricultural sector faces challenges due to poor water retention by the soil, often caused by the erosion of fertile topsoil that holds moisture, resulting in substantial runoff during irrigation [1, 2]. This inefficiency results in significant water and nutrient loss, necessitating more frequent irrigation to sustain crop growth [3]. Additionally, both irrigation and rainfall-induced runoff wash away essential nutrients, reducing soil fertility and ultimately reducing crop yields [4]. Consequently, farmers must invest more in fertilisers and irrigation to compensate for these losses. Intensive agricultural practices, particularly in sloped terrains, exacerbate soil erosion, stripping away nutrient-rich topsoil and increasing sedimentation in water bodies. Additionally, runoff carries herbicides, fertilisers, and other agrochemicals into freshwater systems, contaminating drinking water sources and disrupting aquatic ecosystems. This escalating environmental and public health threat endangers aquatic life and poses risks to humans and animals reliant on these water supplies. As agricultural expansion continues, these challenges are expected to intensify, underscoring the urgent need for innovative solutions to minimise water

runoff, enhance soil retention, and promote sustainable farming practices.

A potential strategy to reduce fertiliser and water loss during irrigation and ultimately improve crop yield is the application of superabsorbent polymers (SAPs) to the soil [5–9]. The addition of SAPs to soil can minimise irrigation frequency due to improved water retention capacity by the soil and reduced fertiliser loss [10–12]. SAPs are macromolecules normally synthesised from polyacrylic acid, polyacrylamides, and polyacrylonitriles, crosslinked with inorganic salt derivatives to form a three-dimensional network [13, 14]. These polymers are stimuli-responsive, enabling the controlled release of absorbed molecules [15, 16], a property that can be exploited for the slow release of water and fertilisers. Incorporating fertilisers into the SAP matrix not only enhances plant growth but also minimises environmental risks associated with conventional fertiliser leaching [17]. However, effective absorption of aqueous solutions requires a highly crosslinked network, which can be achieved through either chemical or physical crosslinking methods. Physical methods involve hydrogen bonding, ionic interactions, hydrophobic bonding, and polymer chain entanglement, allowing SAP formation without the use of a chemical crosslinker [18, 19]. This approach preserves

the natural polymer's low toxicity and biocompatibility, but results in SAPs with reversible crosslinking, rendering them unstable [18]. In contrast, chemical cross-linking forms covalent bonds, producing SAPs with superior structural integrity and durability. Due to their highly cross-linked networks, SAPs can absorb several hundred times their weight in water, making them valuable for use in several industries, from sanitary products to wastewater treatment [19–22]. In agriculture, beyond reducing water run-off, SAPs play a crucial role in lowering water consumption by minimising the need for frequent irrigation, thereby contributing to more sustainable farming practices.

However, a major drawback of most conventional SAPs is their fully synthetic nature, as they are derived entirely from petroleum-based precursors, rendering them non-biodegradable and contributing to long-term environmental pollution [23]. These synthetic SAPs persist in the environment, leading to concerns about plastic waste accumulation. Therefore, the development and use of environmentally friendly, biodegradable SAPs (Bio-SAPs) derived from natural sources are highly desirable to minimise the negative environmental impacts associated with conventional polymers. Although fully bio-based SAPs provide environmental benefits, they often suffer from reduced absorption capacity and weaker structural stability compared to synthetic SAPs. A study by Chen and co-workers synthesised a series of fully natural SAPs and their maximum adsorption capacity was 24 ± 2 g/g, which is significantly lower than that of conventional synthetic SAPs [24]. In another study, carboxymethyl cellulose extracted from cardboards was cross-linked with citric acid to form a fully natural and bio-based SAP. The maximum absorption was found to be 45 g/g [25]. Therefore, semi-synthetic SAPs, composed of naturally occurring polysaccharides and synthetic monomers, offer an ideal alternative and are gaining increasing attention in both research and industrial applications. This is because they combine the desirable properties of bio-SAPs, such as their partial biodegradability, with the superior water absorption and structural integrity typically associated with synthetic SAPs [23, 26–28]. Yassine and co-workers synthesised a semi-synthetic superabsorbent polymer (SAP) using cellulose extracted from banana peels, cross-linked with acrylic acid and itaconic acid. The resulting hybrid SAP exhibited a maximum water absorption capacity of 1240 ± 60 g/g, significantly outperforming fully bio-based SAPs [29]. This performance is comparable to that of conventional synthetic SAPs, highlighting the potential of semi-synthetic formulations to balance high absorption with improved sustainability. Wang and co-workers reported the degradation of a semi-synthetic SAP synthesized from carboxymethyl cellulose and synthetic monomers. Soil burial

experiments demonstrated that the SAP exhibited good biodegradability, with approximately 44% mass loss for 100 days [30].

Polysaccharide-based SAPs offer additional advantages, because the abundance of reactive groups in polysaccharides enables the development of diverse and functional materials [31]. Common sources include alginate, cellulose, chitosan, and starch, with chitin ranking as the second most abundant biopolymer after cellulose. It is a natural polymer found primarily in the exoskeletons of crustaceans (such as crabs and shrimp), insects, and the cell walls of fungi [32–35]. Its wide availability and biodegradability make it an attractive resource for developing environmentally friendly materials such as SAPs for agricultural use.

This study explores the synthesis of two semi-synthetic SAPs derived from chitin extracted from shellfish and acrylic acid as the monomer. Both SAPs were synthesised using aqueous solutions as the solvent, one with deionised water and the other with a urea solution, to enhance chitin solubility and improve the properties of the resulting polymer [36–38]. The study compares the chemical structure, absorption behaviour, and urea slow-release kinetics of the two SAPs. Additionally, these SAPs were loaded with indole butyric acid (IBA), a plant hormone, to assess their effects on plant growth. The effect of pH on zeta potential and the absorption capacity was also examined.

The novelty of this study lies in its direct comparison of chitin-based superabsorbent polymers (SAPs) synthesised with and without urea, assessing structural, physico-chemical, and functional performance. The findings clarify whether urea incorporation delivers measurable agronomic benefits or if a urea-free formulation can achieve comparable results, guiding more sustainable and cost-effective SAP development.

Synthesis

Materials and Methods

All reagents and solvents were purchased from Sigma-Aldrich or Merck and used without further purification. Shellfish chitin powder ($\geq 99\%$ purity), sodium hydroxide (NaOH, analytical grade), acrylic acid (AA, 99%), N, N'-methylenebisacrylamide (MBA, 99%), ammonium persulfate (APS, 98%), urea ($\geq 99\%$), methanol ($\geq 99.5\%$), and ethanol ($\geq 99.5\%$) were obtained from the above suppliers. Deionised water was used in all experiments. These reagents were bought in high purity and were used without further purification.

The synthesised SAPs were characterised using the following techniques. The ATR-FTIR spectra were recorded

in the 400–4000 cm^{-1} range using a Bruker Tensor 27 IR spectrometer with 16 scans per sample at a resolution of 0.5 cm^{-1} . The instrument was equipped with a Pike Miracle single-bounce diamond ATR crystal, and data analysis was performed using OPUS v1.1 software. Scanning Electron Microscopy (SEM) analysis was performed using a JSM-7800 F extreme-resolution analytical field emission instrument. The Malvern Zetasizer was used to obtain the zeta potential of the SAPs at different pH levels at 25 °C. 0.1 M CH_3COOH and 0.1 M NaOH were added to deionised water to adjust the pH between 3 and 11. UV/Vis spectra were recorded using a Varian Cary 50 Conc. ultraviolet/visible spectrophotometer.

Synthesis of Chitin SAPs

For the preparation of the SAPs, two variations were synthesised concurrently, namely Shellfish Chitin-No Urea (SCNU) and Shellfish Chitin-Urea (SCU). The synthesis followed a modified procedure by Liu and co-workers [39]. Chitin powder (1.5 g) was added to an aqueous NaOH solution (prepared by dissolving 8 g of NaOH in 12 g of water) and left at room temperature for 8 h to allow sodium ions to penetrate the chitin structure. The chitin solution then underwent a freeze-thaw process, where it was cooled to -24 °C for 24 h, followed by heating to 40 °C for two hours. This freeze-thawing process was repeated once more. For the SCU SAP, 4 g of urea was dissolved in 73 mL of distilled water and stirred until a clear gel formed, which was then added to the chitin solution. For the SCNU SAP, the same chitin solution as above was prepared and after freeze-thawing, the solution was mixed with 73 mL of pure deionised water, omitting urea from the procedure.

The second part of the synthesis involves a graft polymerisation process, which began with transferring 20 g of each of the two chitin solutions into two separate beakers. This was followed by the addition of the initiator, ammonium persulfate (0.05 g), to each beaker. The mixtures were stirred at room temperature for 30 min to activate the reaction. A crosslinker stock solution was prepared by dissolving *N*, *N*-methylenebisacrylamide (100 mg) in acrylic acid (84 mL). From the stock solution, 8.4 mL was added to each chitin solution, allowing cross-linking between the acrylic acid monomer with the chitin polysaccharide. The reaction mixtures were stirred at 60 °C for two hours, during which a slurry began to form. The reaction was left undisturbed overnight to allow the formation of a solid polymer gel. To clean the resulting SAPs, the polymer gels were soaked in methanol for 8 h and subsequently washed three times with ethanol. Finally, the purified SAPs were left to air dry overnight.

Swelling Behaviour

The water absorption and desorption behaviour of the chitin-based SAPs were evaluated over time using a gravimetric method. Each SAP sample (0.5 g) was pre-weighed and placed in a tea bag before being immersed in 1000 mL of water. The absorption capacity was measured at regular intervals until swelling equilibrium was reached.

Before weighing, the tea bag was removed from the water and allowed to drain excess liquid. To ensure accurate measurement, the surface of the tea bag was lightly dabbed with a dry cloth to remove residual water. A control experiment was also conducted by immersing an empty tea bag in water and following the same procedure. After a predetermined time, the swollen tea bag was weighed.

The swelling rate was calculated using Eq. (4) [40, 41];

$$Q = \frac{m_2 - m_1}{m_1}$$

Where:

Q (g/g) = water absorption capacity

m_1 = mass of dry SAP

m_2 = mass of wet swelled SAP

For desorption studies, the SAPs were removed from the teabags and weighed periodically over 14 days under ambient conditions, with temperatures ranging from 25 to 30 °C.

Biodegradation Assessment via the Soil Burial Method

To evaluate the biodegradability of the SAPs, a laboratory-scale soil burial test was conducted. To facilitate ease of retrieval post-testing, they were placed in permeable fabric pouches, which are also commercial tea bags. These samples were buried at a specific depth, exposing them to microbial activity, as soil serves as a natural habitat for microorganisms. The composting medium ensured enhanced air and water permeability around the specimens. The test was carried out over seven days and in triplicate. Upon completion, the polymer samples were retrieved, thoroughly rinsed with distilled water, and dried. The level of biodegradation was quantified using the determination of sample weight loss at intervals following burial and before burial. The reduction in weight served as an indicator of the extent of biodegradation within the soil environment.

Post-synthetic Urea Loading

Urea (0.5 g, 8.3 mmol) was dissolved in 400 mL of deionised water and stirred to form a homogeneous solution. The solution was divided equally into two parts, and 0.5 g of each SAP sample was immersed in the respective urea solution. The two SAPs were left for 24 h to allow absorption of the aqueous urea solution into the SAP network. After this, the SAPs were left for two weeks for the water molecules to naturally desorb from the SAPs. This was followed by vacuum drying to ensure complete removal of residual moisture. The dried SAPs were then weighed to determine the amount of urea loaded based on the increase in mass. The SCNU SAP showed a final mass of 0.549 g, while the SCU SAP reached 0.602 g, indicating successful urea loading. Urea release was further confirmed using UV/Vis spectroscopy.

Urea Slow Release

The time-dependent urea release behaviour of the SAP upon immersion in water was investigated using UV/Vis spectroscopy [42, 43]. Each SAP was weighed and divided into four equal parts (0.25 g) and placed in separate 500 mL beakers labelled according to the sampling time intervals; 30 min, 1 h, 3 h, 5 h and 24 h. The samples were immersed in water, and at each designated time point, 10 mL of the solution was removed. To each sample, 1 mL of hydrochloric acid and 10 mg (0.067 mmol) of the colour reagent *p*-dimethylbenzaldehyde were added. The samples were left to stand for 10 min to allow the solution to stabilise before analysis.

Plant Growth Effect

The indole butyric acid (IBA) solution was prepared by dissolving 3 mg of IBA in 110 mL of water to be absorbed by 238 mg of SAP. The SAP and acid water were left to be absorbed for 24 h.

Potting soil (ca. 9 g) was blended with the neat and IBA modified SAPs (ca. 5 mg dry material or 3.5 mg wet SAP), and radish sparkler seeds were sown 5 mm deep. The seedlings were initially cultivated in a propagator for seven days before being exposed to atmospheric conditions. Watering was performed weekly, with 5 mL of water administered every seven days. Plant growth was monitored through daily photographic documentation, beginning seven days after planting.

Results and Discussion

Synthesis

The SAPs were synthesised following a modified procedure by Lui et al. [39], involving NaOH pre-treatment and freeze-thaw cycling to enhance chitin modification. Two variations were prepared: SCU, incorporating urea, and SCNU, synthesised without urea. Graft polymerisation was initiated with ammonium persulfate, followed by cross-linking with *N,N*-methylbisacrylamide and acrylic acid at 60 °C, forming a polymer gel. The EDS analysis of SCU confirms the presence of urea as indicated by the detection of nitrogen, while SCNU showed no nitrogen signal (Figure S2 and S3 in the Supplementary Information).

Infrared Spectroscopy

Attenuated Total Reflection Fourier Transform Infrared (ATR-FTIR) spectroscopy was used to compare the functional groups of the starting materials (chitin and acrylic acid) with those of the synthesised SAPs. The FTIR spectra presented in Fig. 1, highlight the key vibrational peaks (labelled 1–5), each corresponding to a specific functional group.

For pure acrylic acid, a broad O-H stretching band appears between 3010 and 3700 cm^{-1} , characteristic of hydroxyl groups engaged in hydrogen bonding [44]. This broad band is also observed in the spectra of pure chitin as well as in both SAPs. In this case, the band is attributed to the overlapping O-H and N-H stretching vibrations arising from hydroxyl groups and amide functionalities [45]. Again, in the spectra of both the SAPs and chitin, an absorption band appears at 2925 cm^{-1} , which is attributed to the C-H stretching vibrations of methylene ($-\text{CH}_2$) groups [46]. This band reflects the presence of aliphatic C-H bonds in the polymer backbones of these materials. This absorbance band is also present in the spectra of pure acrylic acid, but it is slightly shifted to a lower wavenumber due to differences in the chemical environment. In the fingerprint region, the spectrum of acrylic acid displays a strong C=O stretching vibration of carboxylic acids at approximately 1700 cm^{-1} . After SAP formation, this peak shifts to a lower wavenumber and decreases in intensity, indicating the consumption of free carboxylic acid groups and the formation of new ester and amide functional groups. The ester and amide functional groups are represented as amide I (1620 cm^{-1}) and amide II (1550 cm^{-1}). The Amide I is attributed to the C=O stretching and amide II is due to the N-H bending and C-N stretching [47, 48]. In pure chitin, the C=O peak is also slightly shifted compared to free acrylic acid, as the carbonyl groups are part of amide functionalities rather than

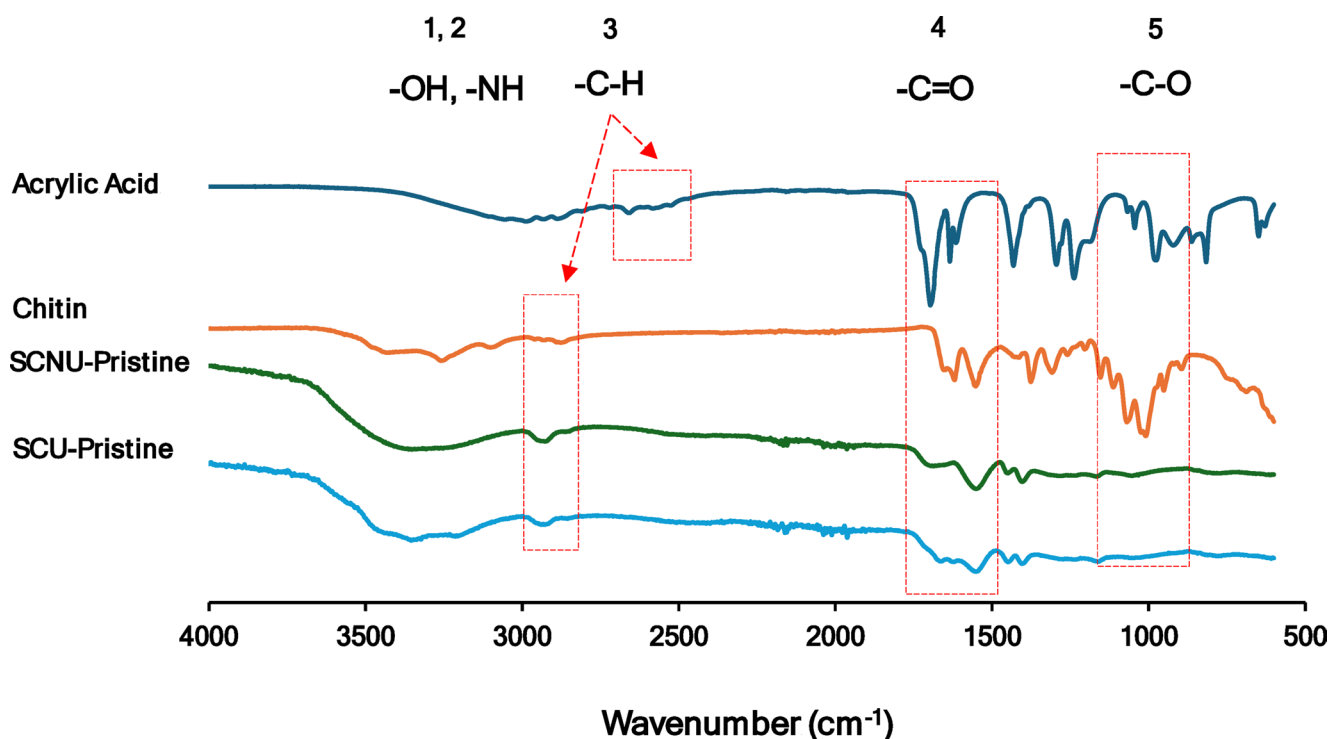


Fig. 1 The FTIR spectra of the monomers and pristine SAPs

free -COOH groups. So pure chitin also has the amide I and II characteristics.

Lastly, the characteristic absorbance band observed at approximately 1000 cm^{-1} in the FTIR spectrum of chitin corresponds to C-O stretching vibrations of hydroxyl (C-OH) and ether (C-O-C) groups within the chitin backbone [47, 48]. This band is prominent in pure chitin but is significantly reduced or absent in the spectra of the superabsorbent polymers (SAPs). During graft copolymerization, the hydroxyl groups of chitin react with the vinyl ($\text{C}=\text{C}$) groups of acrylic acid monomers, leading to covalent modification and the formation of the SAPs. The disappearance of the peak near 1000 cm^{-1} , therefore, indicates the consumption or alteration of these C-O functionalities. Additionally, it has been reported that changes in crystallinity and molecular ordering of the chitin SAPs may influence vibrational modes that cause the loss of this absorption band [49].

Scanning Electron Microscopy and Thermal Analysis

Scanning Electron Microscopy (SEM) was used to examine the surface morphology of the synthesised superabsorbent polymers (SAPs). Figure 2 displays the SEM images of (a) SCU and (b) SCNU, both revealing a distinctly wrinkled surface texture. This morphological feature suggests a high surface area, which could enhance water absorption and retention. Additionally, the wrinkled structure could

also indicate polymer chain interactions that contribute to the mechanical stability of the SAPs. While both variants exhibit similar surface patterns, variation in structural compactness and cross-linking density may influence their respective absorption and release properties [50, 51].

The thermal decomposition profiles of the SAPs are shown in Fig. 3. SCNU exhibits a two-step thermal decomposition process. The first decomposition step occurs at $269\text{ }^{\circ}\text{C}$ with an approximate weight loss of 63% and this step is attributed to the degradation of the polymer backbone, including the chitin and the grafted acrylic acid molecules. The second step occurs at $288\text{ }^{\circ}\text{C}$ with a small weight loss of $\sim 2.5\%$ and it is associated with the charring of the remaining polymer and the decomposition of residual cross-linked fragments and side groups.

SCU, on the other hand, displays a three-step decomposition thermal profile. The first step with a mass loss of 11% occurs between 116 and $224\text{ }^{\circ}\text{C}$, and this step corresponds to the loss of bound water and urea molecules. This assignment aligns with literature-reported urea decomposition, which typically occurs between $132\text{ }^{\circ}\text{C}$ and $200\text{ }^{\circ}\text{C}$ [52, 53]. This initial decomposition step is absent in the urea-free SAP, further supporting this assignment. The second step, occurring at $345\text{ }^{\circ}\text{C}$ with a 9.6% mass loss, is linked to the decomposition of the polymer backbone, including the cleavage of the chitin pyranose rings [39, 54]. The final step, at $428\text{ }^{\circ}\text{C}$ ($\sim 22\%$ loss), is attributed to the breakdown of

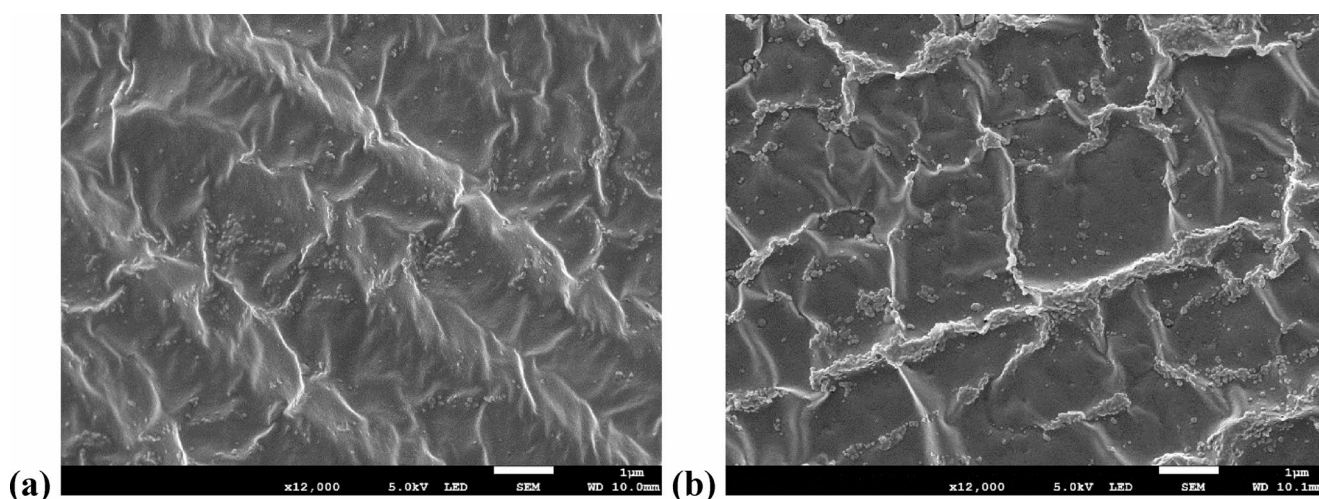


Fig. 2 SEM images of (a)SCU and (b) SCNU

more thermally stable polymer chains and crosslinked fragments within the network.

Overall, SCU demonstrates greater thermal stability than SCNU, indicating that urea incorporation during synthesis enhances the structural integrity of the SAP, likely due to urea-induced crosslinking and reinforcement of the polymer network.

Effect of Zeta Potential

Zeta potential quantifies the electrostatic potential at the interface between a solid surface and a surrounding liquid. When SAPs come in contact with water, they typically acquire a surface charge, resulting in a zeta potential. That charge influences the interactions between the polymer chains and water molecules, ultimately affecting the material's water absorption capacity. The effect of pH on the zeta potential of the SAPs in a water medium was evaluated across a pH range of 3 to 11. Figure 4 presents the zeta potential versus pH curves of the two chitin SAPs, with their corresponding values detailed in the Supporting Information (Table S1). Typically, as pH increases, the zeta potential of a material becomes more negative [55–59], a trend observed for both SAPs. At the lowest pH of 3, the zeta potential for SCNU measured -6.19 mV, decreasing to -40.21 mV at pH 11. A similar trend was observed in SCU SAP, where the zeta potential shifted from -5.59 mV at pH 3 to -43.25 mV at pH 11. These results provide important insights into the surface charge behaviour of the SAPs and their potential impact on their water absorption capacity under varying pH conditions.

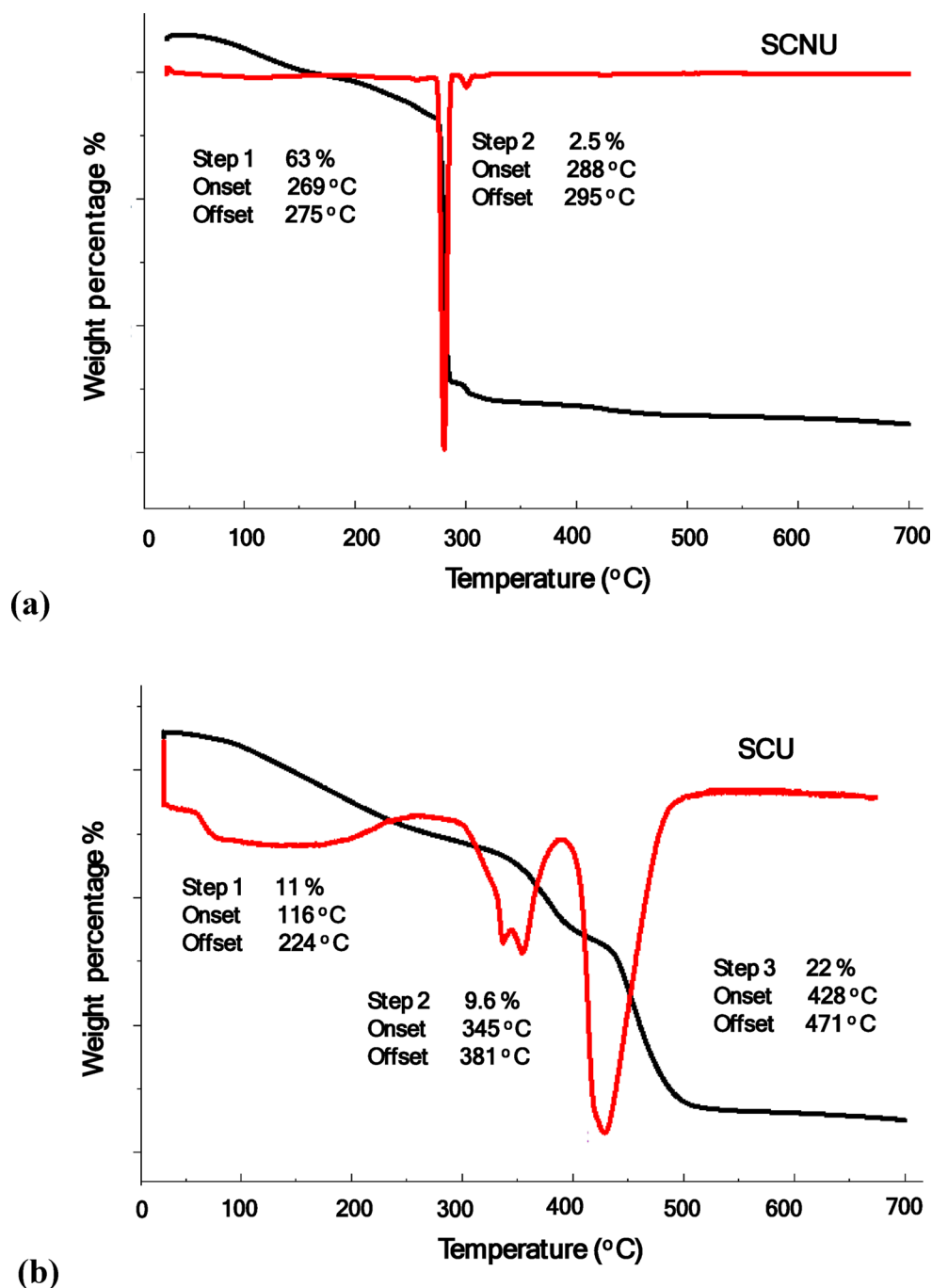
At highly acidic pH levels (1–3), the anionic carboxylate groups in the SAP network become protonated by hydrogen ions (H^+), leading to a slightly higher zeta potential [60]. The zeta potential or surface charge of a material in a

water medium depends on the ionisation of acidic moieties ($COOH$) [56, 60]. At low pH, these acidic groups remain largely protonated and uncharged, reducing the polymer's overall polarity and making it less hydrophilic compared to when these groups are ionised at higher pH.

In acidic environments, dissociated protons contribute to the formation of hydronium ions (H_3O^+), which interact with the SAP. Some hydronium ions protonate the polymer's functional groups, while others remain in the solution, influencing electrostatic interactions at the SAP-water interface. Due to their partial hydrophobic character, hydronium ions tend to accumulate near the polymer surface, adsorbing onto the uncharged polymer chains and imparting a slightly positive surface charge. Consequently, the zeta potential remains only mildly negative, resulting in a relatively low overall surface charge. This reduced surface charge weakens electrostatic repulsion between polymer chains, allowing them to remain more compact. As a result, swelling is restricted, limiting the SAP's water absorption capacity in acidic conditions.

As the pH increases, the concentration of hydronium ions in the surrounding water medium decreases, reducing their adsorption onto the polymeric chains of the SAPs. Additionally, the ionisation of acidic functional groups becomes more pronounced at higher pH levels, contributing to a more negatively charged surface and a progressively more negative zeta potential. This increase in negative charge enhances electrostatic repulsion between the polymer chains, promoting a more expanded structure that facilitates water absorption. This suggests that at pH levels above 4, the SAPs exhibit a higher water absorption capacity compared to more acidic conditions (pH 1–4).

Fig. 3 TGA thermograms of the two pristine SAPs **(a)** SCU and **(b)** SCNU



Effect of pH on Water Absorption

Studies indicate that the pH of a surrounding solution influences the water absorption capacity of SAPs [23, 61, 62]. To assess this effect, we measured the mass of water absorbed per gram of the SAPs at different pH values (3, 5 and 9), as shown in Fig. 5. In highly acidic conditions (pH 1–3), the anionic carboxylate groups within the SAP network become protonated by free hydrogen ions (H^+) [58, 60, 63]. This protonation reduces electrostatic repulsion between the

polymer chains, leading to a more compact and less hydrophilic structure that limits water absorption by the SAP. At pH 3, the SCNU and SCU SAPs exhibited relatively low water absorption capacities of 10 g/g and 50 g/g, respectively. Maximum water absorption for both chitin SAPs was observed at a pH of 5, with the urea SAP (SCU) demonstrating the highest water absorption of 302 g/g, while the non-urea SAP (SCNU) reached 115 g/g. These findings align with the zeta potential measurements, reinforcing the trend that higher pH levels enhance SAP water absorption.

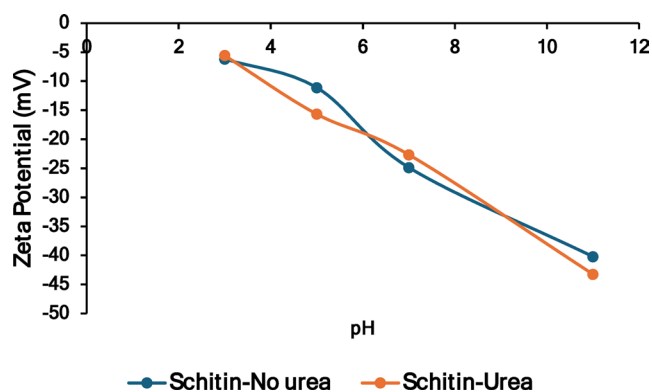


Fig. 4 Graph of zeta potential vs pH for SCNU and SCU SAPs

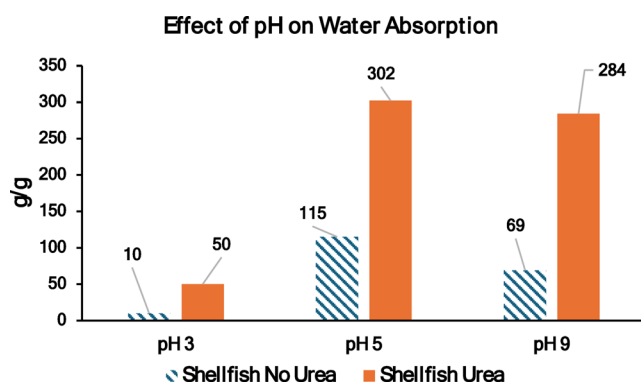
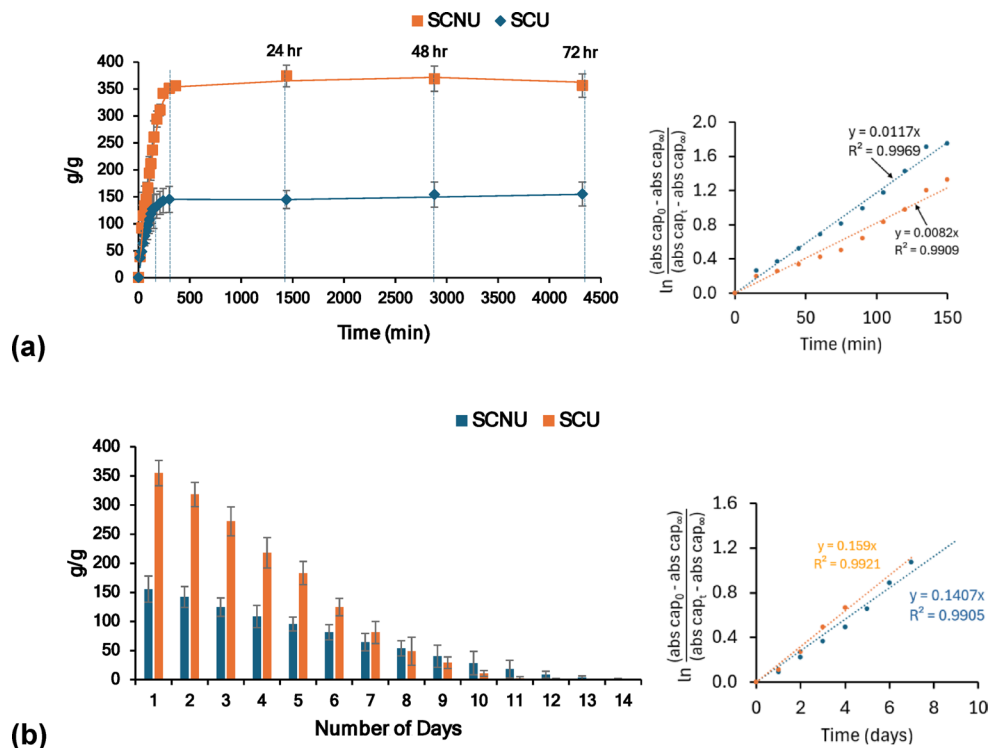


Fig. 5 Effect of pH on water absorption capacity (g/g) of the SAPs

Fig. 6 Comparative analysis of (a) water absorption and (b) desorption behaviour over time for SCU and SCNU SAPs



In addition to pH, the presence of cations (e.g. Na^+ , K^+) in aqueous media also affect the water absorption capacity of the SAPs [64]. In highly basic conditions ($\text{pH} > 8$), these cations interact with the anionic polymer chains, leading to charge shielding and hydrophobic effects that reduce water absorption. These results suggest that pH 5 is the optimal condition for maximum water absorption in the SCU and SCNU SAPs.

Adsorption and Desorption Behaviour

The primary mechanism driving water absorption and swelling in SAPs is osmotic pressure. As water enters the polymer network, the confined carboxylate groups expand, facilitating swelling and enhanced water retention. To assess the absorption kinetics of the SAPs, a gravimetric teabag method was utilised at room temperature, comparing the swelling behaviour of SCU and SCNU (Fig. 6a) [65–69]. The results indicate that SCU exhibited a significantly higher absorption capacity than SCNU, with maximum absorption occurring at 240 min for SCNU (155 g/g) and 360 min for SCU (355 g/g). The incorporation of urea in the SAP enhances the solubility of the chitin starting material, ultimately improving its reactivity and the physical properties of the SAPs, particularly their water absorption capacity. Urea further promotes a more porous polymer network and increases water retention through hydrogen-bond interactions. In addition, it mitigates the hydrophobicity of chitin chains, preventing self-aggregation and increasing

chain mobility, thereby contributing to improved absorption performance [36, 37, 70].

Adsorption kinetics were done at pH 5, as the SAPs demonstrated maximum uptake at that pH value. The maximum water absorption capacities of SCU and SCNU were compared with other efficient SAPs, including synthetic, semi-synthetic, and natural types (Table 1).

The semi-synthetic chitin-based SAPs developed in this work provide a balanced performance in comparison to fully natural SAPs, which typically exhibit lower adsorption capacities (40–46 g/g), and fully synthetic SAPs, which achieve high capacities (up to 1690 g/g) but rely fully on non-biodegradable, petroleum-based precursors. As semi-synthetic SAPs, SCU and SCNU exhibit substantially higher absorption than natural counterparts while offering improved biodegradability and environmental compatibility compared to fully synthetic SAPs.

The desorption behaviour of the SAPs was evaluated over 14 days at 25 °C to evaluate their ability to retain and gradually release moisture. The results indicate a steady release of water molecules over time, suggesting that these SAPs could potentially reduce irrigation frequency when incorporated into the soil. As shown in Fig. 6b, the SAPs progressively release water and return to their original dry mass after approximately 11 days, demonstrating their capacity for sustained moisture retention. These findings highlight the potential of chitin-based SAPs in improving water management strategies for agricultural applications.

Table 1 Comparison of the maximum water adsorption capacity (g/g) of the SAPs developed in this study with other efficient SAPs (natural, synthetic, and semi-synthetic types) reported in the literature in deionised water

SAP	Category	(g/g)	Ref
SAP-85 (CA-MSC-GC)	Natural	40	[24], (2025)
CMC-CA (Citric acid)	Natural	46	[25], (2019)
Semi IPN PVA/poly(IA-co-AMPS)	Synthetic	60	[71], (2023)
SCNU	Semi-synthetic	155	This work
Cellulose-Co-AMPS	Semi-synthetic	280	[72], (2021)
SAP2 (CMC/starch/citric acid)	Natural	287	[73], (2023)
SCU	Semi-synthetic	355	This work
Starch-g-poly(AA-co-AM)/PVA/clinoptilolite	Semi-synthetic	365	[74], (2018)
PAA	Synthetic	395	[75], (2023)
SA-DAC (Cellulose)	Semi-synthetic	1240	[29], (2024)
P(AA-co-AMPS)/PVP	Synthetic	1690	[76], (2024)

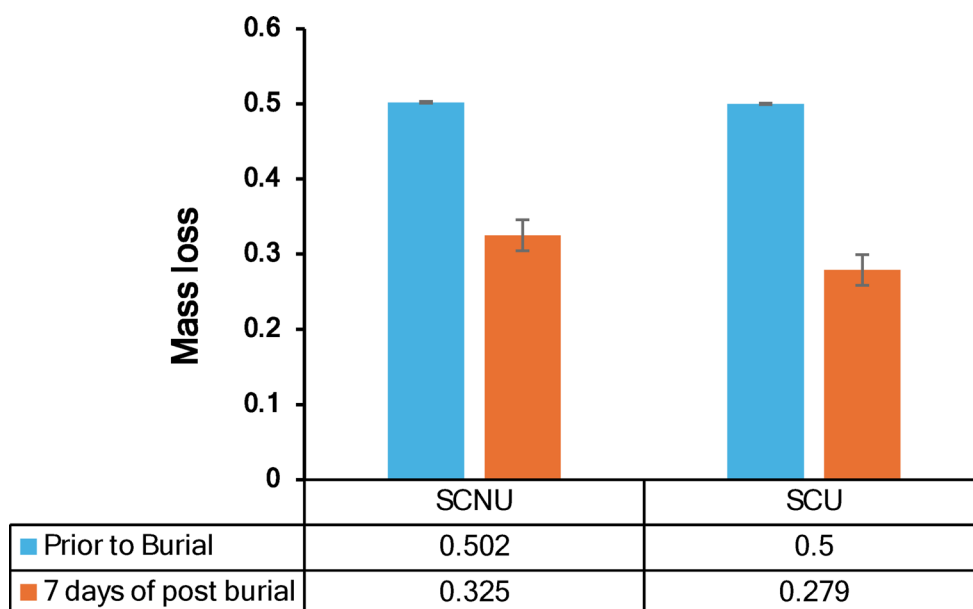
Pseudo-first order kinetics was applied to both the absorption and desorption data. The apparent first-order rate constants, k_{abs} , were determined as 0.0117 min^{-1} for SCNU and 0.0082 min^{-1} for SCU, indicating that the SAP that does not contain urea absorbs water at a faster rate than the SAP containing urea (SCU). Although SCU exhibits a lower k value than SCNU, indicating slower initial water uptake, it ultimately achieves a much higher maximum absorption capacity (355 g/g vs. 155 g/g). The swelling behaviour of the SAPs can be characterised by their water absorption kinetics, which often follow Fickian or non-Fickian diffusion mechanisms. Fickian swelling is controlled primarily by the diffusion of water molecules into the polymer matrix, where the rate of diffusion is much slower than the relaxation of the polymer chains. In contrast, non-Fickian (or anomalous) swelling occurs when the rates of diffusion and polymer chain relaxation are comparable, leading to more complex swelling kinetics. The relatively consistent rate constants and the fitting to pseudo-first-order kinetics suggest that the swelling process is dominated by Fickian diffusion, where water diffusion governs the absorption rate, and polymer relaxation occurs sufficiently quickly to not limit the process.

As for the desorption kinetics, the apparent first-order rate constants, k_{desorp} , were determined to be 0.1407 day^{-1} for SCNU and 0.159 day^{-1} for SCU. Desorption is thus ca. 75–120 times slower than absorption for SCNU and SCU, respectively. This is considered advantageous since the slow release of water is preferred for the intended use.

Biodegradation of SAPs Using the Burial Method

To assess the biodegradability of the SAPs, a laboratory-scale soil burial test was performed under controlled conditions. The SAP burial test results (Fig. 7) show a decrease in mass for both SAPs after 7 days, indicating microbial and environmental degradation. SCNU's mass decreased from 0.502 g to 0.325 g (35.3% loss), while SCU's mass decreased from 0.500 g to 0.279 g (44.2% loss). The slightly higher mass loss for SCU suggests greater susceptibility to biodegradation, likely due to its more open and porous structure from urea incorporation during synthesis. This enhanced porosity facilitates microbial access and penetration, accelerating degradation compared to the denser SCNU network. Hence, these findings highlight the benefits of semi-synthetic SAPs as eco-friendly alternatives to fully synthetic SAPs, offering improved biodegradability without compromising performance.

Fig. 7 Mass loss of SCNU and SCU SAPs over 7 days in soil burial tests, indicating biodegradability



Urea Release

To detect urea in the SAPs, an adaptation of Ehrlich's reaction was used. In this method, *p*-dimethylaminobenzaldehyde (*p*-DMAB) reacts with urea in the presence of hydrochloric acid to produce a yellow-green compound. The intensity of the resulting colour, which correlates directly with the urea concentration, was measured using UV/Vis spectroscopy, with absorbance observed at approximately 430 nm [77].

Initially, both pristine SCNU and SCU were analysed for urea content using UV/Vis spectroscopy. As expected, SCNU showed no characteristic urea absorbance peak, since urea was not incorporated during its synthesis, whereas SCU displayed a clear peak. Figure 8 shows the UV/Vis spectra of the pristine SAPs (after 24 h swelling in distilled water), with an absorption peak at 430 nm for SCU and no corresponding peak for SCNU. Similarly to the EDS results, this further confirms the presence of urea in SCU. This indicates that the NaOH/urea solution not only enhances the absorption capacity of the SAP but also incorporates urea molecules into the SAP structure, potentially serving as a supplementary fertiliser for plants.

Thereafter, both SAPs were post-synthetically treated with urea to determine whether urea could be incorporated into the SAP network after synthesis. Following treatment, the SAPs were washed with methanol and evacuated under vacuum to remove any surface-bound urea. UV/Vis spectroscopy was used to monitor the urea release from the SAPs over 24 h, ultimately revealing that SCNU still lacked the characteristic 430 nm peak, indicating that post-synthetic urea loading was unsuccessful (Figure S1). In contrast, SCU showed an increase in absorbance after loading, suggesting some additional urea uptake. Prior to post-synthetic

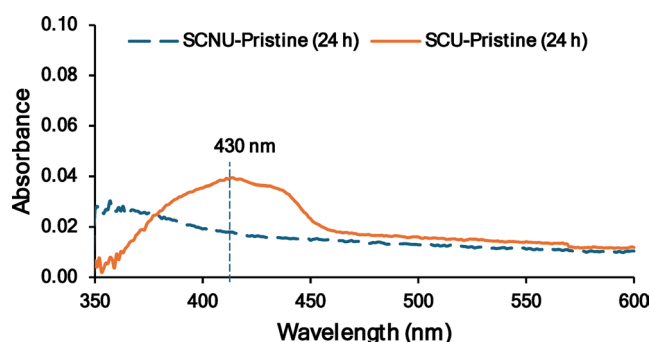
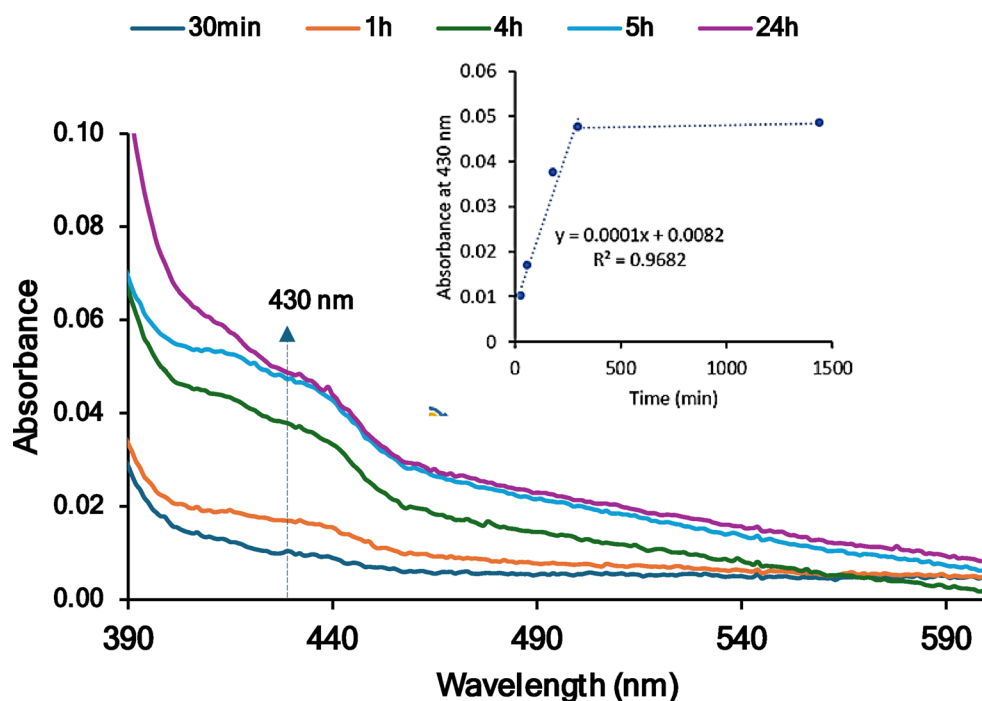


Fig. 8 UV/Vis spectra for detecting the presence of urea in the synthesised SAPs

modification (PSM) loading, SCU's maximum absorbance peak was 0.04 A (Fig. 8), which increased to 0.07 A after PSM loading (Fig. 9). The observed modest increase indicates successful urea loading, albeit to a limited extent under the current experimental conditions. However, similar to the enhanced water uptake by SCU, the higher urea uptake observed in SCU (and not in SCNU) can be attributed to its synthesis method, in which urea was present during polymerisation. Urea helps create a more porous polymer network and enhances the retention of aqueous solutions through hydrogen bonding. It also reduces the hydrophobicity of chitin chains, preventing them from aggregating and increasing chain mobility, which further improves the uptake of water and dissolved molecules [37].

Since SCNU could not be successfully loaded, only the urea release kinetics of SCU are presented in Fig. 9. The slow release of urea into distilled water was monitored and maximum release was reached within 24 hours. The results indicate urea is gradually released from SCU following a zero-order mechanism with the apparent rate constant $k_0' =$

Fig. 9 Urea slow-release behaviour of SCU in deionised water, as measured by UV/Vis spectroscopy over 24 hours



0.0001 min^{-1} . Urea release did not increase after 5 h, implying that all the additional absorbed urea was released from SCU.

Plant Growth

Germination and plant growth assays of SCU on radish seedlings were carried out under controlled laboratory conditions using a plant-growing light for 14 days. Plants were watered on days 1, 8, and 14, and photographic records from days 5, 7, and 14 are presented in **Fig. 10**. By day 5, seeds grown in soil mixed with SCU or indole butyric acid-enriched SCU germinated more quickly than those in neat soil (without SAP). On day 7, seedlings appeared healthy in all treatments; however, radishes grown in indole butyric acid-enriched SCU exhibited exceptionally vigorous root growth, with roots extending beyond the pot (**Figure S4**). By day 14 (seven days after the previous watering), seedlings in neat soil had wilted and failed to recover, whereas those in SCU-treated and IBA-SCU-treated soils remained healthy, growing beyond the limits of their seedling pots and even overlapping into neighbouring pots. This suggests that SCU gradually releases water, sustaining seedling health for a longer period compared to neat soil.

Conclusion

This study demonstrates the potential of semi-synthetic chitin-based superabsorbent polymers (SAPs) as sustainable and multifunctional materials for agriculture. Urea-containing SAPs (SCU) exhibited superior water absorption, reaching 355 g/g within, compared to 155 g/g for non-urea SAPs (SCNU), while both materials showed rapid biodegradability within 7 days. This improvement is attributed to the enhanced solubility of the starting materials (during synthesis) in the NaOH/urea solvent system, which facilitates better polymer network formation. Desorption studies showed that the SAPs gradually release water, extending soil moisture and potentially reducing irrigation frequency. SCU also promoted plant growth, as seedlings grown in SCU-enriched soil remained healthy without additional watering. These findings highlight the dual function of SCU in water retention and nutrient delivery, supporting the development of eco-friendly semi-synthetic SAPs for sustainable agricultural practices.

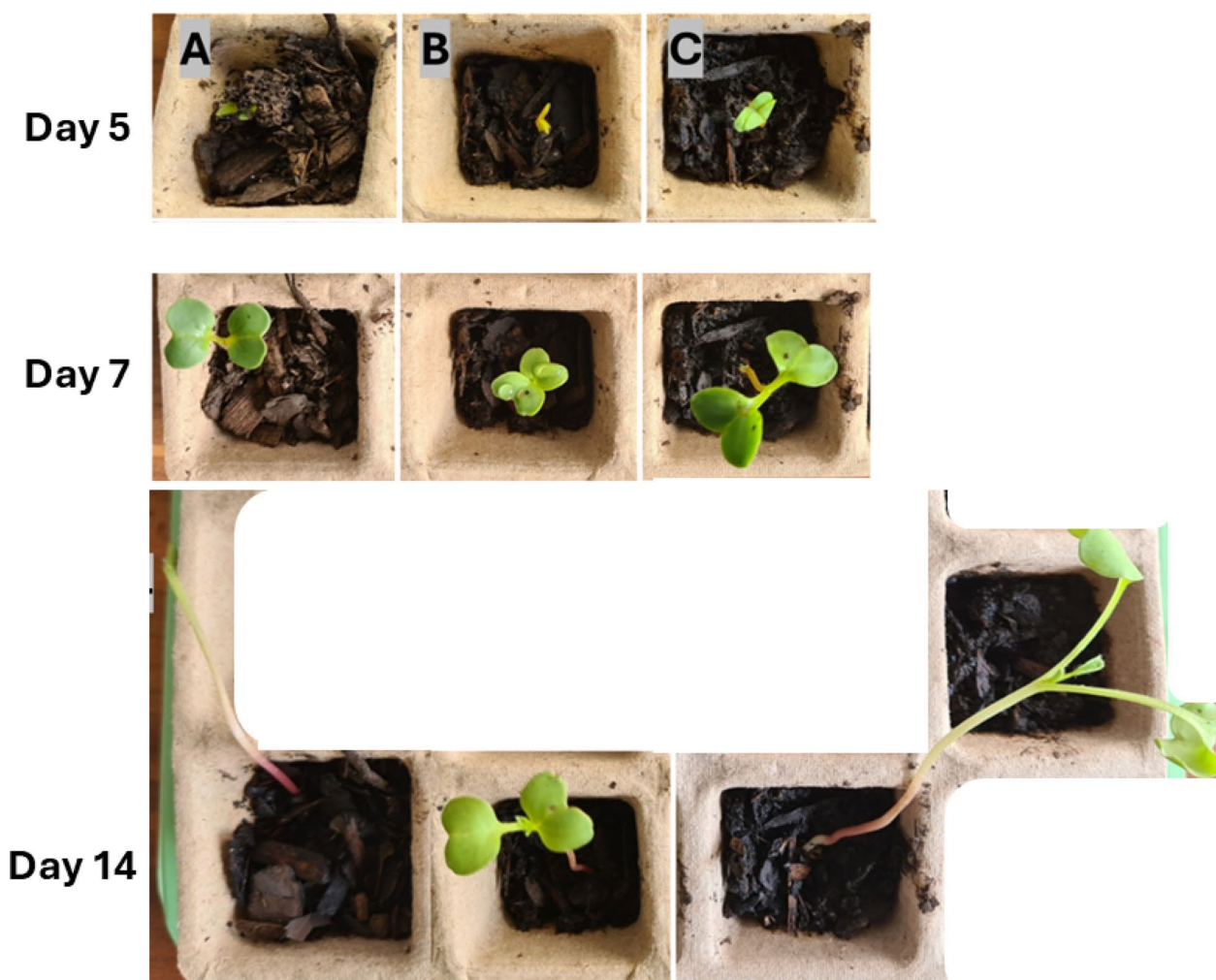


Fig. 10 Photos of radish seedlings planted in (A) untreated soil, (B) soil mixed with SCU, and (C) soil mixed with SCU enriched with indole butyric acid at days 5, 7, and 14 after planting

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Data Availability The data presented in this study are available on request from the corresponding author. The data are not publicly available due to the copyright and intellectual property policies of the University of the Free State.

Declarations

Competing interests The authors declare no competing interests.

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